School of Engineering

E211 – Operations Planning II

Lesson 4 Worksheet

<u>Warm up question (pick 1)</u>

Below are some real-world queuing systems. State the arrival, queue and service characteristics in each queuing system.

a. Food ordering

<u>Arrival:</u>
Population size: Infinite
Arrival pattern: Random
Behaviour: Depends on the queue (Mostly customers are patient, some might be impatient)

<u>Queue:</u>
Queue capacity: Unlimited
Queue configuration: Multiple line due to multiple counters
Queue discipline: First-Come-First-Serve

<u>Service:</u>
Service time pattern: Random, Exponential
Number of servers: Multiple
Number of phases: 2 (After order placement → collect food from counter)

b. Car refuelling at petrol station

c. Security check at airport

d. Queuing at a theme park ride.

School of Engineering

School of Engineering

Problem Statement

Figure 1 shows the self-checkout area at an IKEA store.

Part 1

Figure 1: IKEA self-checkout area

Data gathered:

- Arrival rate = 150 customers per hour
- Mean checkout time = 2.5 minutes

a. Which queuing model should you use to study the IKEA self-checkout area? State the assumptions.
M/M/K (M/M/8).

Assumptions:
Input source: Infinite population; no balks, no reneging
Arrival process: Number of arrivals follows Poisson distribution
Queue: Unlimited; single line
Queue discipline: FIFO (FCFS)
Service mechanism: Multiple servers, single phase system
Service time: Negative exponential distribution
Arrivals and service times are independent of the number of customers in system
Arrival rate must be smaller than sum of service rate of servers, i.e. λ/kμ

b. Determine the minimum number of checkout machines needed in order for the queuing system to be in steady state.

$\lambda < K\mu$ **(Arrival rate < No of servers*Service Rate)**

Arrival rate (λ) = 150/60 = 2.5/min
Service rate (μ) = 1 / 2.5 = 0.4/min
2.5 < k(0.4)
2.5 < 0.4k
k = 2.5/0.4 = 6.25 = 7 (minimum number)

c. Assume there are 8 checkout machines. Determine the performance measures of the queuing system – utilization, waiting time, time in system, queue length and number in system.

Utilization **(ρ)** = 0.7813
Waiting time **(Wq)** = 0.60
Time in system **(Ws)** = 3.10
Queue length **(Lq)** = 1.49
Number in system **(Ls)** = 7.74

Problem Statement:

Arrival rate = 150 customers per hour
Mean checkout time = 2.5 minutes

M/M/k

Arrival rate	λ	2.500	per minute
Service rate	μ	0.4	per minute
No. of servers	k	8	
Utilisation	ρ	0.7813	
Prob (no customers)	P_0	1.58E-03	
No. of customers in queue	Lq	1.49	
No. of customers in system	Ls	7.74	
Waiting time in queue	Wq	0.60	minutes
Time in system	Ws	3.10	minutes

d. Calculate the probability that at least 1 self-checkout machine is available at any time.

No of servers (k) = 8

Probability = P0 + P1 + P2 + P3 + P4 + P5 + P6 + P7
= 0.581781247

No. of customers in system, n	$\frac{(\lambda/\mu)^n}{n!}$	Probability of n customers in system, Pn	
		$\frac{(\lambda/\mu)^n}{n!}P_0$ for $0 \le n \le k$	$\frac{(\lambda/\mu)^n}{k!k^{n-k}}P_0$ for $n > k$
0	1	0.00158428	0.659221436
1	6.25	0.00990175	0.515016747
2	19.53125	0.030942969	0.402356834
3	40.69010417	0.06446452	0.314341276
4	63.57828776	0.100725812	0.245579122
5	79.4728597	0.125907265	0.191858689
6	82.78422885	0.131153401	0.149889601
7	73.91449005	0.117101251	0.117101251

e. Calculate the probability that there are at least 2 customers in the queue.
No of servers (k) = 8

Probability = 1 – (P0 + P1 + P2 + P3 + P4 + P5 + P6 + P7 + P8 + P9)
= 0.255260469

Since n (9) > k (8), we take the value of **0.071472931** when calculating.

No. of customers in system, n	$\frac{(\lambda/\mu)^n}{n!}$	Probability of n customers in system, Pn	
		$\frac{(\lambda/\mu)^n}{n!}P_0$ for $0 \le n \le k$	$\frac{(\lambda/\mu)^n}{k!k^{n-k}}P_0$ for $n > k$
0	1	0.00158428	0.659221436
1	6.25	0.00990175	0.515016747
2	19.53125	0.030942969	0.402356834
3	40.69010417	0.06446452	0.314341276
4	63.57828776	0.100725812	0.245579122
5	79.4728597	0.125907265	0.191858689
6	82.78422885	0.131153401	0.149889601
7	73.91449005	0.117101251	0.117101251
8	57.74569535	0.091485352	0.091485352
9	40.10117733	0.063531495	0.071472931

<u>Part 2</u>

Suppose that there is a cashier counter set up to assist customers who are unable to perform self-checkout. Your task is to determine an average service time standard for the cashier checkout so that the service level and staff utilization are reasonable. Initial estimates are that on average, 20 customers are expected to use the cashier checkout every hour.

Discuss the approach you would take to carry out your task.
I will use M/M/1.

Arrival rate = 20/60 = 0.333/min

Assume system utilization = 0.8,
System utilization = Arrival rate/Service rate
System utilization = 0.333/service rate
0.8 = 0.333 / service rate
Service rate = [(20/60) / 0.8)] = 0.41666

Wq (average time in queue) = 9.601, Ws (average time in system) = 12.001
Add more counters to reduce waiting times.

School of Engineering

E211 – Operations Planning II

Lesson 16 Worksheet

Shortest Path Problem (Dijkstra's algorithm)

1. Referring to distribution routes below (Figure 1), draw a network diagram of the possible delivery routes from Johor Bahru to each of the other cities. What do the **nodes** represent in this network? What do the **arcs** represent in the network?

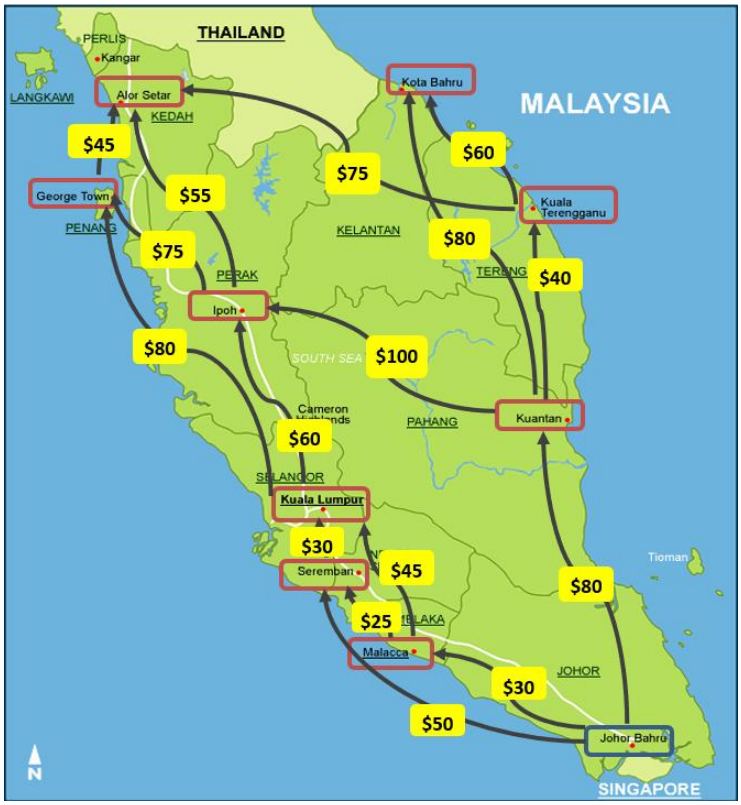


Figure 1: WantTheToys Unit Cost Distribution Routes in Malaysia

Nodes represent the cities in this network. Arcs represent the delivery route cost of delivering toys from Johor Bahru (JB) to 9 other cities.

2. List down all possible routes that can transport the toys from Johor Bahru to Alor Setar. How many possible routes are there? () represent the total cost to deliver

   Johor Bahru – Malacca – Kuala Lumpur – George Town – Alor Setar (200)

   Johor Bahru – Malacca – Kuala Lumpur – Ipoh – George Town – Alor Setar (255)

   Johor Bahru – Malacca – Kuala Lumpur – Ipoh – Alor Setar (190)

   Johor Bahru – Malacca – Seremban – Kuala Lumpur – George Town – Alor Setar (210)

Johor Bahru – Malacca – Seremban – Kuala Lumpur – Ipoh – George Town – Alor Setar (265)
Johor Bahru – Malacca – Seremban – Kuala Lumpur – Ipoh – Alor Setar (200)

Johor Bahru – Seremban – Kuala Lumpur – George Town – Alor Setar (205)
Johor Bahru - Seremban – Kuala Lumpur – Ipoh – George Town – Alor Setar (260)
Johor Bahru – Seremban – Kuala Lumpur – Ipoh – Alor Setar (195)

Johor Bahru – Kuantan – Ipoh – George Town – Alor Setar (300)
Johor Bahru – Kuantan – Ipoh – Alor Setar (235)
Johor Bahru – Kuantan – Kuala Terengganu – Kota Bahru (180)
Johor Bahru – Kuantan – Kuala Terengganu – Alor Setar (195)
Johor Bahru – Kuantan – Kota Bahru (160)

There are possible 14 routes.

3. Out of all the possible routes from Johor Bahru to Alor Setar, which route do you think has the **least** cost for delivering one truckload of toys? What type of problem is this and what is one approach you can use to solve this problem?
   Least cost: Johor Bahru – Kuantan – Kota Bahru (160).
   This is **shortest path problem** and one approach to solve this problem is using **Dijkstra's algorithm.**

4. Applying Dijkstra's algorithm, work out the delivery routes that would achieve the least cost for delivering the toys **from Johor Bahru to each of the other 9 cities**. You may use the following template to work out your solution.

AY 2025 Sem 2 E211
L16 Templates.xlsx

Iteration	Solution Set	Arcs	Total Cost
1	JB	**JB - M** JB - S JB - K	**30** 50 80

Iteration	Solution Set	Arcs	Total Cost
2	JB M (30)	**JB - S** JB - K M - S M - KL	**50** 80 30+25=55 30+45=75

Iteration	Solution Set	Arcs	Total Cost
3	JB	JB - K	80
	M (30)	**M - KL**	**30+45=75**
	S (50)	S - KL	50+30=80

Iteration	Solution Set	Arcs	Total Cost
4	JB	**JB - K**	**80**
	M (30)	KL - GT	75+80=155
	S (50)	KL - IPOH	75+60=135
	KL (75)		

Iteration	Solution Set	Arcs	Total Cost
5	JB	K - IPOH	80+100=180
	M (30)	**K - KT**	**80+40=120**
	S (50)	KL - GT	75+80=155
	KL (75)	KL - IPOH	75+60=135
	K (80)	K - KB	80+80=160

Iteration	Solution Set	Arcs	Total Cost
6	JB	KL - GT	75+80=155
	M (30)	**KL - IPOH**	**75+60=135**
	S (50)	K - IPOH	80+100=180
	KL (75)	KT - KB	120+60=180
	K (80)	KT - AS	120+75=195
	KT (120)	K - KB	80+80=160

Iteration	Solution Set	Arcs	Total Cost
7	JB	K - KB	80+80=160
	M (30)	IPOH - GT	135+75=210
	S (50)	IPOH - AS	135+55=190
	KL (75)	**KL - GT**	**75+80=155**
	K (80)	KT - KB	120+60=180
	KT (120)	KT - AS	120+75=195
	IPOH (135)		

Iteration	Solution Set	Arcs	Total Cost
8	JB	GT - AS	155+45=200
	M (30)	IPOH - AS	135+55=190
	S (50)	KT - AS	120+75=195
	KL (75)	KT - KB	120+60=180
	K (80)	**K - KB**	**80+80=160**

	KT (120) IPOH (135) GT (155)		

Iteration	Solution Set	Arcs	Total Cost
9	JB M (30) S (50) KL (75) K (80) KT (120) IPOH (135) GT (155) KB (160)	GT - AS **IPOH - AS** KT - AS	155+45=200 **135+55=190** 120+75=195

Iteration	Solution Set	Arcs	Total Cost
10	JB M (30) S (50) KL (75) K (80) KT (120) IPOH (135) GT (155) KB (160) AS (190)		

SHORTEST PATH SOLUTION		
From JB to	Shortest Path	Total Cost
M	JB -> M	30
S	JB -> S	50
KL	JB -> M -> KL	75
K	JB -> K	80
KT	JB -> K -> KT	120
IPOH	JB -> M -> KL -> IPOH	135
GT	JB -> M -> KL -> GT	155
KB	JB -> K -> KB	160
AS	JB -> M -> KL -> IPOH -> AS	190

5. What are some assumptions made when deriving the solution to the shortest path problem?
   Total cost of shipment is proportional to the amount shipped
   The shipment follows a single, continuous path from origin to destination

Unit cost of shipment is fixed.

Assumptions: (From 6P Solution)

- Unit cost of shipment does not change with shipment quantity.
- Routes in the network are fixed.
- There is no limitation of route capacity.

Maximum Flow Problem (Ford-Fulkerson's algorithm)

6. Refer to Figure 2, draw the network diagram of the toy delivery routes clearly indicating the Origin and Destination nodes. Indicate the flow capacity along each arc.

Figure 2: WantTheToys Distribution Routes Capacity in Malaysia

Pick any delivery route from Johor Bahru to Alor Setar. What is the maximum quantity of toys (number of truckloads) that can be delivered along that route?

Johor Bahru – Malacca – Kuala Lumpur – George Town – Alor Setar

Max qty: 50

7. Using Ford-Fulkerson's algorithm, determine the maximum quantity of toys (in number of truckloads) that WantTheToys can deliver from Johor Bahru to Alor Setar. You may make use of the template in question 4.

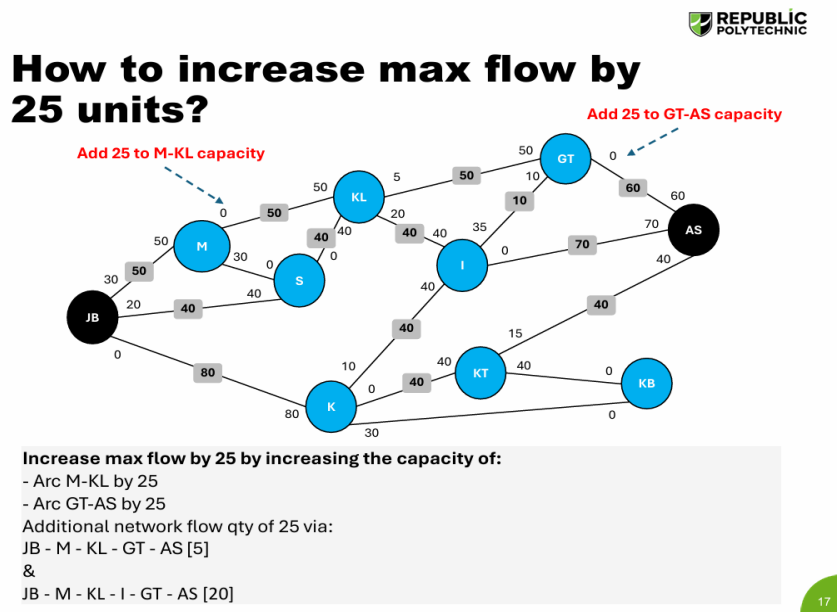
Route	Flow Quantity
JB - M - KL - GT - AS	50
JB - S - KL - I - AS	40
JB - K - KT - KB	40
JB - K - KB	30
JB - K - I - GT - AS	10
Total	**170**

8. What are some assumptions when deriving the solution to the maximum flow problem?
   Cost of delivery is not a factor in determining the routes
   Products are only delivered from Johor to Alor Setar
   Delivery routes and capacity are fixed

9. There is a request for WantTheToys to deliver another 25 truckloads of toys from Johor Bahru to Alor Setar in addition to the max quantity determined above. Determine where the extra delivery capacity should be added and the quantity to add.
   Arc M-KL by 25
   Arc GT-AS by 25

   JB - M - KL - GT - AS [5], JB - M - KL - I - GT - AS [20]

   Answer Key:



10. Would the algorithms you apply to solve the shortest path and maximum flow problems today – Dijkstra's algorithm and Ford-Fulkerson's algorithm – always give optimal solutions?

    Both algorithms give optimal solutions.

School of Engineering

REPUBLIC POLYTECHNIC

E211 – Operations Planning II

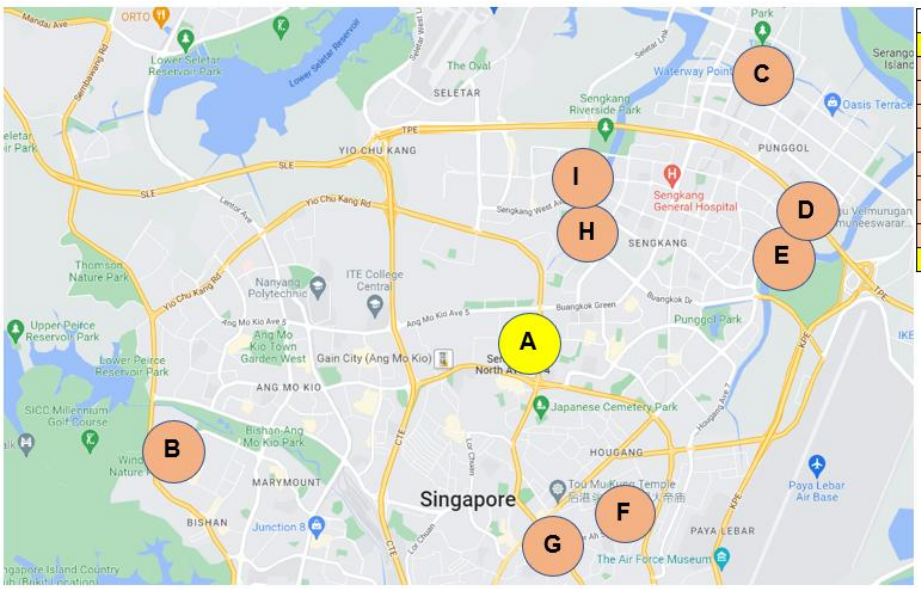
Lesson 18 Worksheet

Problem Statement

WantTheToys Pte Ltd is a Singapore based company that produces educational toys. To carry out the toy distribution in Singapore, WantTheToys is working with ExpressD, a last-mile delivery company.

Every day, ExpressD will deliver products from WantTheToys' warehouse to fulfil customer orders. Drivers often spend time planning their delivery routes as customer locations change daily. To complicate matters, the quantity of products to deliver would sometimes exceed capacity of the van. Also, some customers might request deliveries to be made at specific times. Is there a way to help ExpressD plan their delivery routes efficiently?

Use the data below for the exercise.



Label	Address
A (Depot)	Serangoon North
B	Jalan Angklong
C	Sumang link
D	Rivervale Crescent
E	Rivervale Street
F	Jalan Pelikat
G	Lorong Lew Lian
H	Fernvale Link
I	Sengkang West Way
A (Depot)	Serangoon North

Figure 1: Depot and Delivery Locations

School of Engineering

REPUBLIC POLYTECHNIC

Table 1: Distance in Kilometers Between Locations

From		To (Distance in km)								
		A	B	C	D	E	F	G	H	I
Serangoon North (Depot)	**A**	0	8.1	6	6	5.2	4.3	3.8	4	4
Jalan Angklong	**B**	8.1	0	13.5	14.4	14.3	9.1	6.7	9.5	9.3
Sumang link	**C**	6	13.5	0	3.7	3.6	9.1	9.7	6.2	5.6
Rivervale Crescent	**D**	6	14.4	3.7	0	1.7	6.8	7.1	4.4	4.2
Rivervale Street	**E**	5.2	14.3	3.6	1.7	0	6.5	7.1	4.2	4.5
Jalan Pelikat	**F**	4.3	9.1	9.1	6.8	6.5	0	2.3	7.7	8.2
Lorong Lew Lian	**G**	3.8	6.7	9.7	7.1	7.1	2.3	0	7.2	6.6
Fernvale Link	**H**	4	9.5	6.2	4.4	4.2	7.7	7.2	0	1.3
Sengkang West Way	**I**	4	9.3	5.6	4.2	4.5	8.2	6.6	1.3	0

Table 2: Customer Demand

Customer	Delivery Load (kg)
B	100
C	100
D	200
E	300
F	80
G	70
H	180
I	70

1. Determine a route for the deliveries to the 8 customers using the Nearest Neighbour method.
   Route (tour): A – G – F – E – D – C – I – H – B – A

Route (tour):

From	To	Distance
A	G	3.8
G	F	2.3
F	E	6.5
E	D	1.7
D	C	3.7
C	I	5.6
I	H	1.3
H	B	9.5
B	A	8.1
	Total Dist.	**42.5 km**

School of Engineering

2. Determine a route for the deliveries using Clarke and Wright Savings method. You may use the attached template for workings.

AY 2025 Sem 2 E211
L18 Templates.xlsx

Rank	Saving	From	To	Tour	Remarks
1	9.5	D	E	**(1) A-D-E-A**	Initiate new tour
2	8.3	C	D	**(1) A-C-D-E-A**	Join C and D
3	7.6	C	E		C and E are already assigned to the same route
4	6.7	H	I	**(1) A-C-D-E-A** **(2) A-H-I-A**	Initiate new tour
5	5.8	D	I		D is not adjacent to A
6	5.8	F	G	**(1) A-C-D-E-A** **(2) A-H-I-A** **(3) A-F-G-A**	Initiate new tour
7	5.6	D	H		D is not adjacent to A
8	5.2	B	G	**(1) A-C-D-E-A** **(2) A-H-I-A** **(3) A-F-G-B-A**	Join B and G
9	5.0	E	H	**(1) A-C-D-E-H-I-A** **(2) A-F-G-B-A**	Join E and H; Merge the 2 tours
10	4.7	E	I		E and I are already assigned to the same route
11	4.4	C	I		C and I are already assigned to the same route
12	3.8	C	H		C and H are already assigned to the same route
13	3.5	D	F		D is not adjacent to A
14	3.3	B	F		B and F are already assigned to the same route
15	3.0	E	F		E is not adjacent to A
16	2.8	B	I	**(1) A-C-D-E-H-I-B-G-F-A**	Join B and I; Merge the 2 tours
17	2.7	D	G		D and G are already assigned to the same route
18	2.6	B	H		B and H are already assigned to the same route
19	1.9	E	G		E and G are already assigned to the same route
20	1.2	C	F		C and F are already assigned to the same route
21	1.2	G	I		G and I are already assigned to the same route
22	0.6	B	C		B and C are already assigned to the same route
23	0.6	F	H		F and H are already assigned to the same route
24	0.6	G	H		G and H are already assigned to the same route
25	0.1	C	G		C and G are already assigned to the same route
26	0.1	F	I		F and I are already assigned to the same route
27	-0.3	B	D		B and D are already assigned to the same route
28	-1.0	B	E		B and E are already assigned to the same route

Route (tour): A – C – D – E – H – I – B – G – F – A

From	To	Distance
A	C	6
C	D	3.7
D	E	1.7
E	H	4.2
H	I	1.3
I	B	9.3
B	G	6.7
G	F	2.3
F	A	4.3

Total Dist.	**39.5**	**km**

3. Compare the routes derived from both methods above. Which is a better route? In general, will any of these methods always provide optimal routing solutions?
   Clarke and Wright Savings is a better route. These methods do not guarantee optimal routing solutions.

4. Suppose ExpressD would like to assign 2 vans to provide delivery to the 8 customers. Discuss the approach you could use to plan the assignment of van and the route for each van to take.
   A-C-D-E-H-I-A
   A-F-G-B-A

5. Capacity of ExpressD's vans is **600 kg**. Given the delivery load for each customer below, answer the following questions:

Customer	Delivery Load (kg)
B	150
C	100
D	150
E	200
F	80
G	100
H	180
I	90

a. Are the routes that you worked out previously still feasible? Why?
   (Nearest Neighbour Method):
   Route (tour): A – G – F – E – D – C – I – H – B – A
   100+80+200+150+100+90+180+150=1050

   (Clarke & Wright Savings Method):
   Route (tour): A – C – D – E – H – I – B – G – F – A
   100+150+200+180+90+150+100+80=1050

   The routes are not feasible as total delivery load (1050kg) > capacity (600kg)

b. How should the company plan the delivery routes now? Work out a new solution using Clark and Wright savings method.
   The company should consider the constraints of total delivery load of each customer and total van capacity. Total load should not exceed total capacity.

School of Engineering

Rank	Saving	From	To	Tour	Cumulative Load	Remarks
1	9.5	D	E	**(1) A-D-E-A**	**150+200=350**	Initiate a new route: A-D-E-A
2	8.3	C	D	**(1) A-C-D-E-A**	**100+150+200=450**	Link C with D
3	7.6	C	E			C and E are already assigned to the same route
4	6.7	H	I	**(1) A-C-D-E-A** **(2) A-H-I-A**	**100+150+200=450** **180+90 = 270**	Initiate another route: A-H-I-A
5	5.8	D	I			D is not adjacent to A
6	5.8	F	G	**(1) A-C-D-E-A** **(2) A-H-I-A** **(3) A-F-G-A**	**100+150+200=450** **180+90 = 270** **80+100=180**	Initiate another route: A-F-G-A
7	5.6	D	H			D is not adjacent to A
8	5.2	B	G	**(1) A-C-D-E-A** **(2) A-H-I-A** **(3) A-F-G-B-A**	**100+150+200=450** **180+90 = 270** **80+100+150=330**	Link B with G
9	5.0	E	H	**(1) A-C-D-E-I-A** **(2) A-H-A** **(3) A-F-G-B-A**	**100+150+200+90=540, 180** **80+100+150=330**	Cannot link E with H, because truck capacity (600 kg) will be violated, start new route for H.
10	4.7	E	I			Both E and I are already in a route
11	4.4	C	I			Both C and I are already in a route
12	3.8	C	H			Cannot merge C with H, because truck capacity (600 kg) will be violated
13	3.5	D	F			D is not adjacent to A
14	3.3	B	F			Both B and F are already in a route
15	3.0	E	F			E is not adjacent to A
16	2.8	B	I	**CWS Original route:** **A-C-D-E-H-I-B-G-F-A** **New route (With Vehicle Capacity):** **(1) A-C-D-E-A** **(2) A-H-I-B-G-F-A**	**100+150+200=450,** **180+90+150+100+80=600**	Link B and I
17	2.7	D	G			
18	2.6	B	H			
19	1.9	E	G			
20	1.2	C	F			
21	1.2	G	I			
22	0.6	B	C			
23	0.6	F	H			
24	0.6	G	H			
25	0.1	C	G			
26	0.1	F	I			
27	-0.3	B	D			
28	-1.0	B	E			
Truck 1: A-C-D-E-A with load 450kg and total distance 16.6km **Truck 2: A-H-I-B-G-F-A with load 600kg and total distance 27.9km**						

Truck 1

From	To	Distance
A	C	6
C	D	3.7
D	E	1.7
E	A	5.2
	Total Dist.	**16.6** **km**

Truck 2

From	To	Distance
A	H	4
H	I	1.3
I	B	9.3
B	G	6.7
G	F	2.3
F	A	4.3
	Total Dist.	**27.9** **km**

Routes (tour):

Truck 1: A-C-D-E-A

Truck 2: A-H-I-B-G-F-A

6. If some customers request for the delivery to be done at certain time periods, how should ExpressD consider this in the planning of the route?
   ExpressD should consider the constraints of customers:
   - Time windows (timing open for receiving of goods)
   - Priority
   - Pickup and delivery

7. Discuss the challenges and issues that may cause delivery failure or unexpected delays in deliveries.
   Delivery failures often occur because the address is incorrect, incomplete, or ambiguous. This requires redelivery attempts or returns to the depot, further delaying the process. Shortages of drivers, warehouse staff, or other operational personnel, especially during peak seasons, can slow down sorting, loading, and deliveries.

School of Engineering

REPUBLIC POLYTECHNIC

E211 – Operations Planning II

Quiz 1 Question for Lesson 9 E-Learning

Minimum Cost Flow

WonderToys has manufacturing centers located in Singapore, Bangkok and Manila. WonderToys is working closely with its regional partners and the plan is to ship the toys to customers in the other ASEAN countries – Jakarta, Phnom Penh, Naypyidaw, Hanoi, Kuala Lumpur, Vientiane except Bandar Seri Begawan. The latter is a transshipment node in its supply chain. Figure 1 shows the planned distribution network and Table 1 & Table 2 the respective supply and demand information.

How can WonderToys plan a distribution plan that meets the demand for its educational toys and minimize the total cost of shipment?

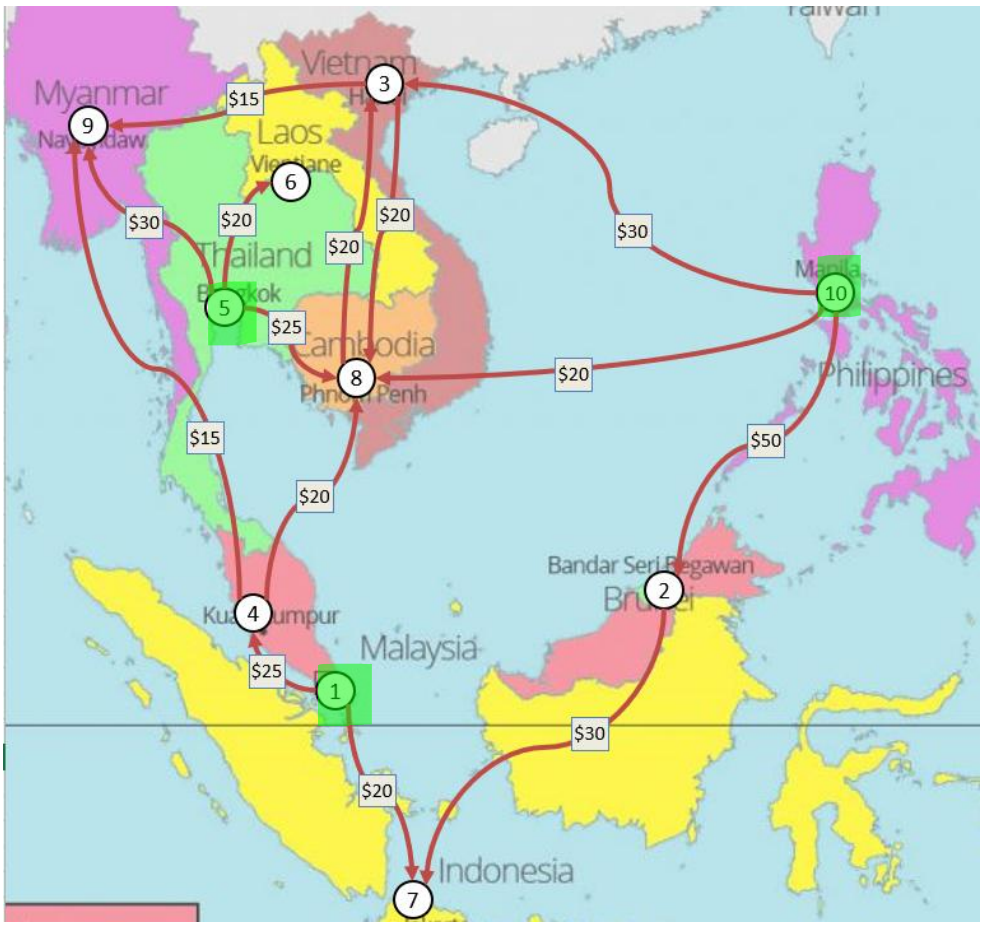


Figure 1: Toy Distribution Network

Table 1: Supply Quantities

Manufacturing Plants	Supply Units
[1] Singapore	**1200**
[5] Bangkok	**800**
[10] Manila	**1000**
Total	**3000**

Table 2: Demand Quantities

Transhipment/ Demand Centres	Demand Units
[2] Bandar Seri Begawan	**0**
[3] Hanoi	**600**
[4] Kuala Lumpur	**800**
[6] Vientiane	**300**
[7] Jakarta	**600**
[8] Phnom Penh	**400**
[9] Naypyidaw	**300**
Total	**3000**

1) How do you define the decision variables used in the LP formulation? In total, how many decision variables are there?
   Answer: Let X_{ij} be the *number of units of toys (in '000s)* to be shipped from node i to node j. In total, there are 14 decision variables:
   X_{14}, X_{17}, X_{27}, X_{38}, X_{39}, X_{48}, X_{49}, X_{56}, X_{58}, X_{59}, X_{83}, X_{102}, X_{103}, X_{108}

2) Write down the objective function of the minimum cost flow problem using the decision variables defined earlier.
   Minimize total shipment cost, $Z = 25X_{14} + 20X_{17} + 30X_{27} + 20X_{38} + 15X_{39} + 20X_{48} + 15X_{49} + 20X_{56} + 25X_{58} + 30X_{59} + 20X_{83} + 50X_{102} + 30X_{103} + 20X_{108}$

3) What are the types of constraints in this model? ***Select all answers that apply.***
   Net flow constraints, non-negativity constraints, upper-bound constraints

4) Match the net flow constraint of each city with its formulation

Supply Quantities:

Manufacturing Plants	Supply Units
[1] Singapore	1200
[5] Bangkok	800
[10] Manila	1000
Total	3000

Demand Quantities:

Transshipment/ Demand Centers	Demand Units
[2] Bandar Seri Begawan	0
[3] Hanoi	600
[4] Kuala Lumpur	800
[6] Vientiane	300
[7] Jakarta	600
[8] Phnom Penh	400
[9] Naypyidaw	300
Total	3000

Supply Node 10: X102 + X103 + X108 = 1000	Net flow constraint for supply from Manila
Demand Node 7: - X17 - X27 = -600	Net flow constraint for demand from Jakarta
Demand Node 4: X48 + X49 -X14 = -800	Net flow constraint for demand from Kuala Lumpur
Transshipment Node 2: X27 - X102 = 0	Net flow constraint for transshipment city Bandar Seri Begawan
Supply Node 1: X14 + X17 = 1200	Net flow constraint for supply from Singapore

5) What is the upper-bound constraint in this network? *Note: There is a maximum shipping capacity of 150 from Bangkok to Naypyidaw*
   Arc 5 to 9: X_{59} **<=** 150

6) Formulate the non-negativity constraint.
   All X_{ij} >= 0

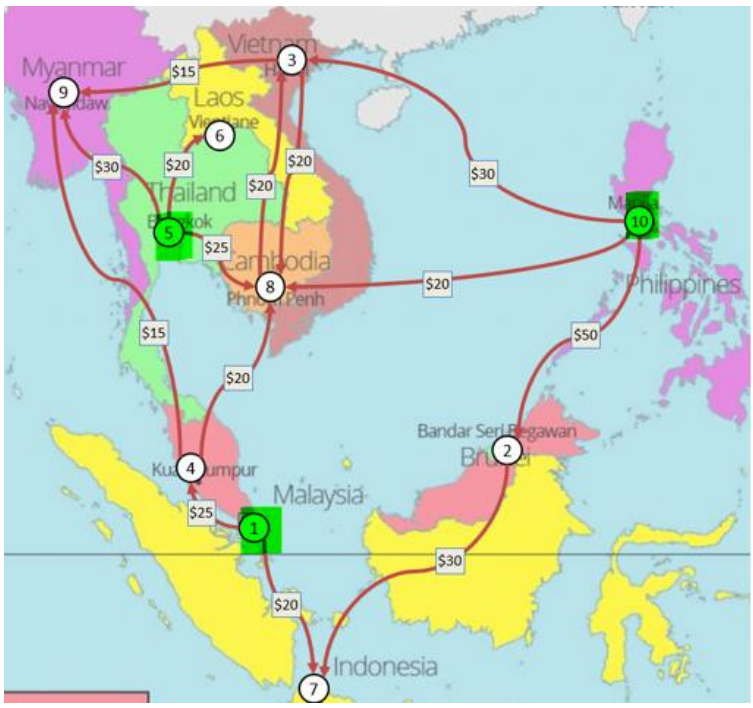
7) How would the constraints in your minimum cost flow LP model change if the plant in Singapore is able to supply 1500 units of toys instead of 1200? ***Select all answers that apply.***
   Note: There is a maximum shipping capacity of 150 from Bangkok to Naypyidaw.

Table 1: Supply Quantities

Manufacturing Plants	Supply Units
[1] Singapore	1500
[5] Bangkok	800
[10] Manila	1000
Total	3000

Table 2: Demand Quantities

Transhipment/ Demand Centres	Demand Units
[2] Bandar Seri Begawan	0
[3] Hanoi	600
[4] Kuala Lumpur	800
[6] Vientiane	300
[7] Jakarta	600
[8] Phnom Penh	400
[9] Naypyidaw	300
Total	3000



Supply Node 1: $X_{14} + X_{17} <= 1500$
Supply Node 5: $X_{56} + X_{58} + X_{59} <= 800$
Supply Node 10: $X_{102} + X_{103} + X_{108} <= 1000$

8) How would the constraints in your minimum cost flow LP model change if the demand at Hanoi has increased to 700? ***Select all answers that apply***.
Note: There is a maximum shipping capacity of 150 from Bangkok to Naypyidaw.

Demand Node 3: $X_{38} + X_{39} - X_{83} - X_{103} >= -700$
Demand Node 4: $X_{48} + X_{49} - X_{14} >= -800$

Demand Node 6: $-X_{56} >= -300$
Demand Node 7: $-X_{17} - X_{27} >= -600$
Demand Node 8: X_{83} (Flow out) $- X_{38} - X_{48} - X_{58} - X_{108}$ (Flow in) $>= -400$
Demand Node 9: $-X_{39} - X_{49} - X_{59} >= -300$

- [x] Demand Node 7: $-X_{17} - X_{27} >= -600$
- [x] Demand Node 3: $X_{38} + X_{39} - X_{83} - X_{103} >= -700$
- [ ] Demand Node 4: $X_{48} + X_{49} - X_{14} >= 800$
- [x] Demand Node 9: $-X_{39} - X_{49} - X_{59} >= -300$
- [x] Demand Node 4: $X_{48} + X_{49} - X_{14} >= -800$
- [x] Demand Node 6: $-X_{56} >= -300$
- [ ] Demand Node 9: $X_{39} + X_{49} + X_{59} = -300$
- [x] Demand Node 8: $X_{83} - X_{38} - X_{48} - X_{58} - X_{108} >= -400$

School of Engineering

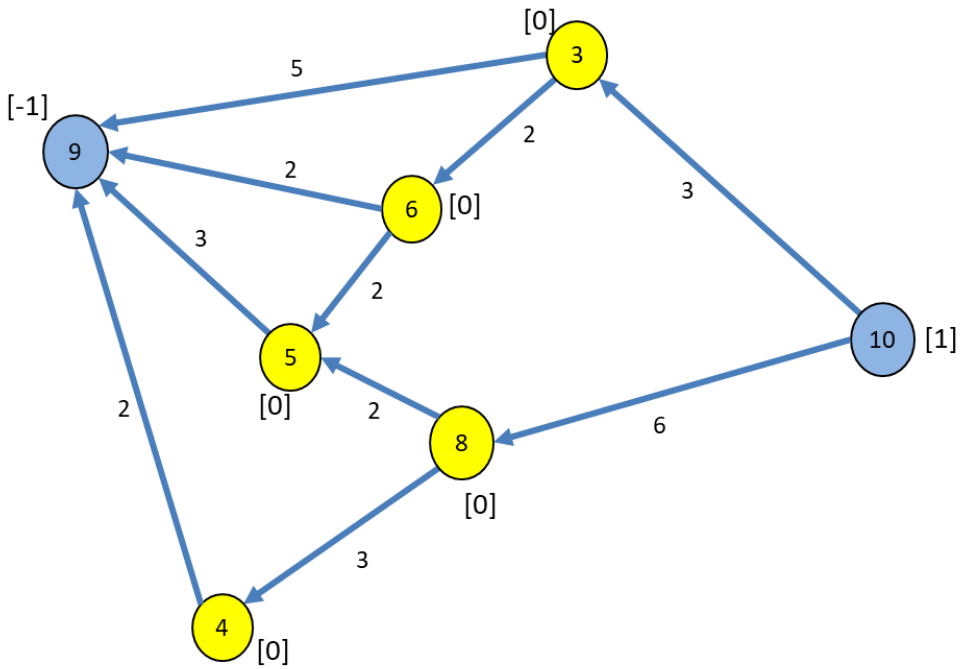
REPUBLIC POLYTECHNIC

E211 – Operations Planning II

Quiz 2 Question for Lesson 9 E-Learning

<u>Shortest Path (LP approach)</u>

Due to sudden demand increasing in Naypyidaw, WonderToys needs to quickly send a shipment of toys from Manila (10) to Naypyidaw (9) next month. The shipment has to pass through Vientiane (6) for quality inspection purposes. Figure below shows the possible shipment routes and travel time along each route (in hours).



1) Determine the respective net flow at each node.

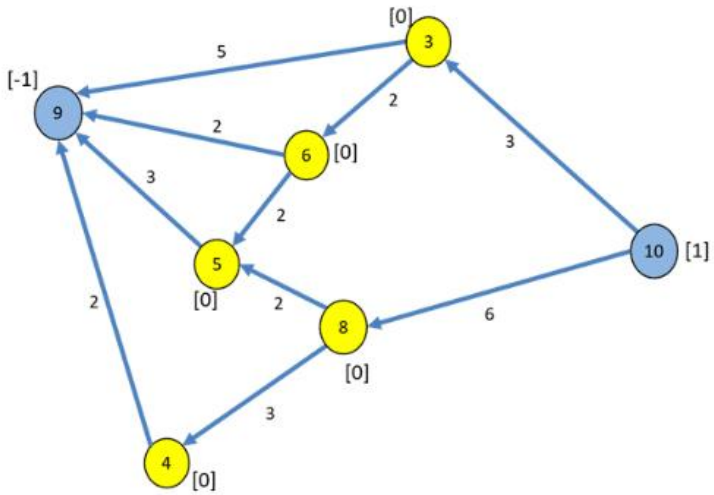
≡	Source is a supply node with	+1 unit of net flow (node 10)
≡	Destination is the demand note with	-1 unit of net flow (node 9)
≡	Transshipment is nodes with	0 unit of net flow (all other locations)

2) Formulate the decision variables. Select all answers that apply.
   - Let X_{ij} be the route from node i to node j
   - In total, there are 10 decision variables: X_{36}, X_{39}, X_{49}, X_{59}, X_{65}, X_{69}, X_{84}, X_{85}, X_{103}, X_{108}
   - X_{ij} is an binary variable

3) Write down the objective function for the shortest path LP model.
   Minimize total traveling time, $Z = 2X_{36} + 5X_{39} + 2X_{49} + 3X_{59} + 2X_{65} + 2X_{69} + 3X_{84} + 2X_{85} + 3X_{103} + 6X_{108}$

4) Construct the net flow constraints. ***Select all answers that apply.***
   Node 3: $X_{36} + X_{39} - X_{103} = 0$
   Node 4: $X_{49} - X_{84} = 0$
   Node 5: $X_{59} - X_{65} - X_{85} = 0$
   Node 6: $X_{65} + X_{69} - X_{36} = 0$
   Node 8: $X_{84} + X_{85} - X_{108} = 0$
   Node 9: $-X_{39} - X_{49} - X_{59} - X_{69} = -1$ (destination)
   Node 10: $X_{103} + X_{108} = 1$ (source)

5) Formulate Mandatory Pass constraint for Vientiane (Node 6).
   Select **two formulas**, one using outflow and the other using inflow arc(s).

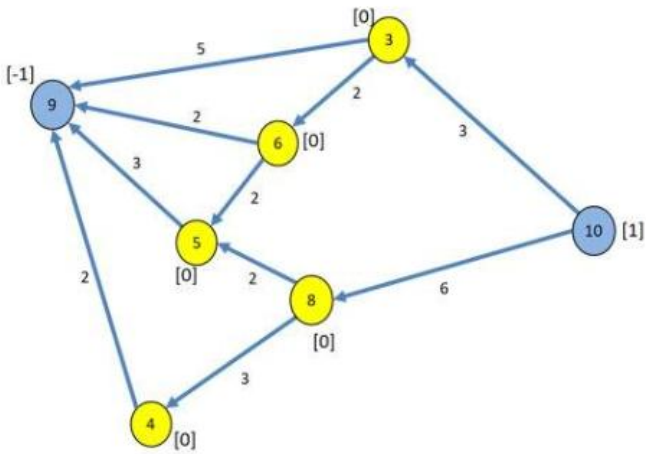


   Node 6: $X_{36} >= 1$
   Node 6: $X_{65} + X_{69} >= 1$

6) Construct non-negativity constraint
   All $X_{ij} >= 0$

7) The picture shows you the solution of the shortest path problem using Excel Solver. From the result, the route is

Decision Variables	Solution	Time (hrs)
x36	1	2
x39	0	5
x49	0	2
x59	0	3
x65	0	2
x69	1	2
x84	0	3
x85	0	2
x103	1	3
x108	0	6



Node 10 to node 3 to node 6 to node 9

8) From the result, the total traveling time is _____ hours.
(2*1)+(2*1)+(3*1) = 7 hours

School of Engineering

REPUBLIC POLYTECHNIC

E211 – Operations Planning II

Quiz 3 Question for Lesson 9 E-Learning

Delivery Routing

As an Operations Manager in a logistics firm, Danny needs to plan the delivery routes according to the customer's locations and demands. On a particular day, Danny needs to formulate a plan to deliver a certain product (packed in boxes) from warehouse (A) to six of its customers. The following table illustrates the distances between the warehouse (A) and the 6 customers (B to G) and their respective demands (in number of boxes).

From	To (Distance in km)							
	A	B	C	D	E	F	G	Demand
A	-	26	14	30	28	29	31	-
B	26	-	15	23	25	8	13	37
C	14	15	-	14	16	15	11	20
D	30	23	14	-	18	21	35	15
E	28	25	16	18	-	33	23	23
F	29	8	15	21	33	-	9	36
G	31	13	11	35	23	9	-	32

Determine a delivery route using the Nearest Neighbour method.

A-C-G-F-B-D-E-A

What is the total distance of the route?
14+11+9+8+23+18+28=111

Can you use another node (e.g. node B) as the starting node in the Nearest Neighbour method?
Yes, however a different solution may result

What if the vehicle can only carry 90 boxes of the product? *Select all answers that apply.*

- One route would be A -> C -> G -> F -> A
- Another route would be A-> B -> D -> E -> A
- The solution would contain 2 separate routes

  A-C-G-F-A (20+32+36=88)
  A-B-D-E-A (37+15+23=75)

School of Engineering

E211 – Operations Planning II

Lesson 20 Worksheet

WantTheToys has manufacturing centers located in Singapore, Bangkok and Manila. The company plans to ship a new batch of toys from the manufacturing centers to customers in other ASEAN countries – Jakarta, Phnom Penh, Naypyidaw, Hanoi, Kuala Lumpur and Vientiane. Bandar Seri Begawan has no demand but serves as a transshipment node in the supply chain. The distribution network as well as the supply and demand information at each node is given below.

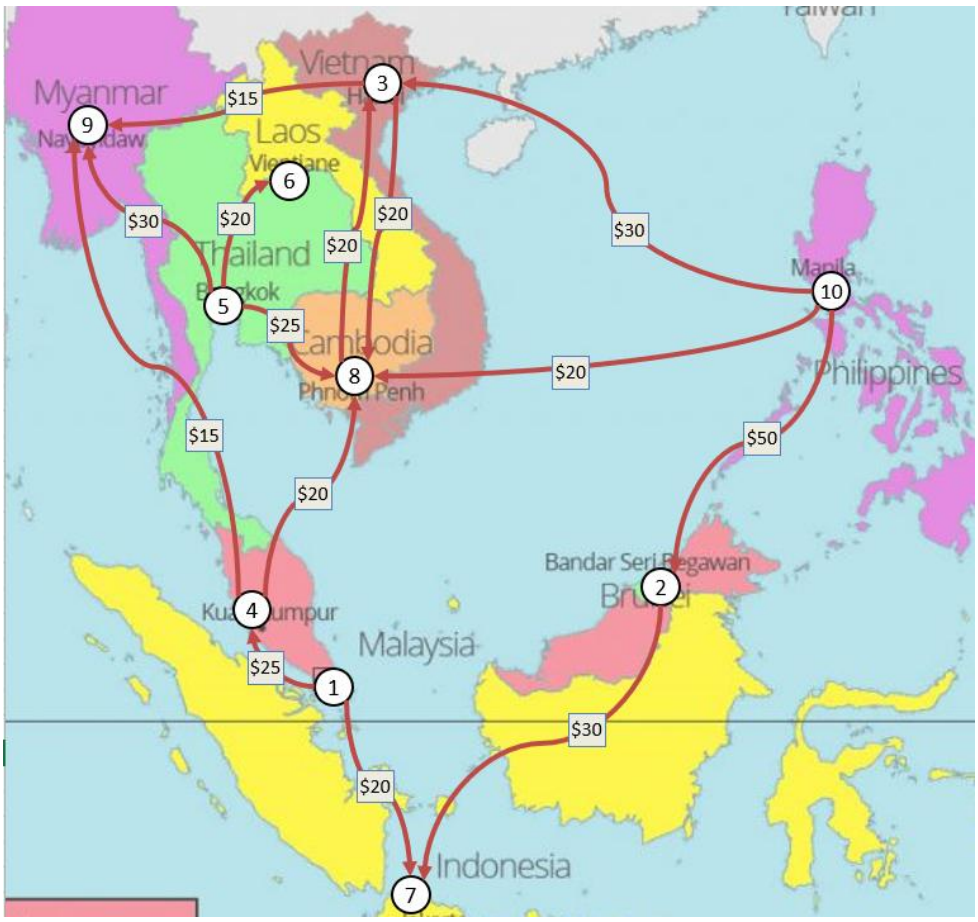


Figure 1: Toy Distribution Network

Table 1: Supply Quantities

Manufacturing Plants	Supply Units
[1] Singapore	**1200**
[5] Bangkok	**800**
[10] Manila	**1000**
Total	**3000**

Table 2: Demand Quantities

Transhipment/ Demand Centres	Demand Units
[2] Bandar Seri Begawan	**0**
[3] Hanoi	**600**
[4] Kuala Lumpur	**800**
[6] Vientiane	**300**
[7] Jakarta	**500**
[8] Phnom Penh	**500**
[9] Naypyidaw	**300**
Total	**3000**

Furthermore, there is a shipping capacity limit of 500 from Manila to Hanoi and 200 from Bangkok to Naypyidaw.

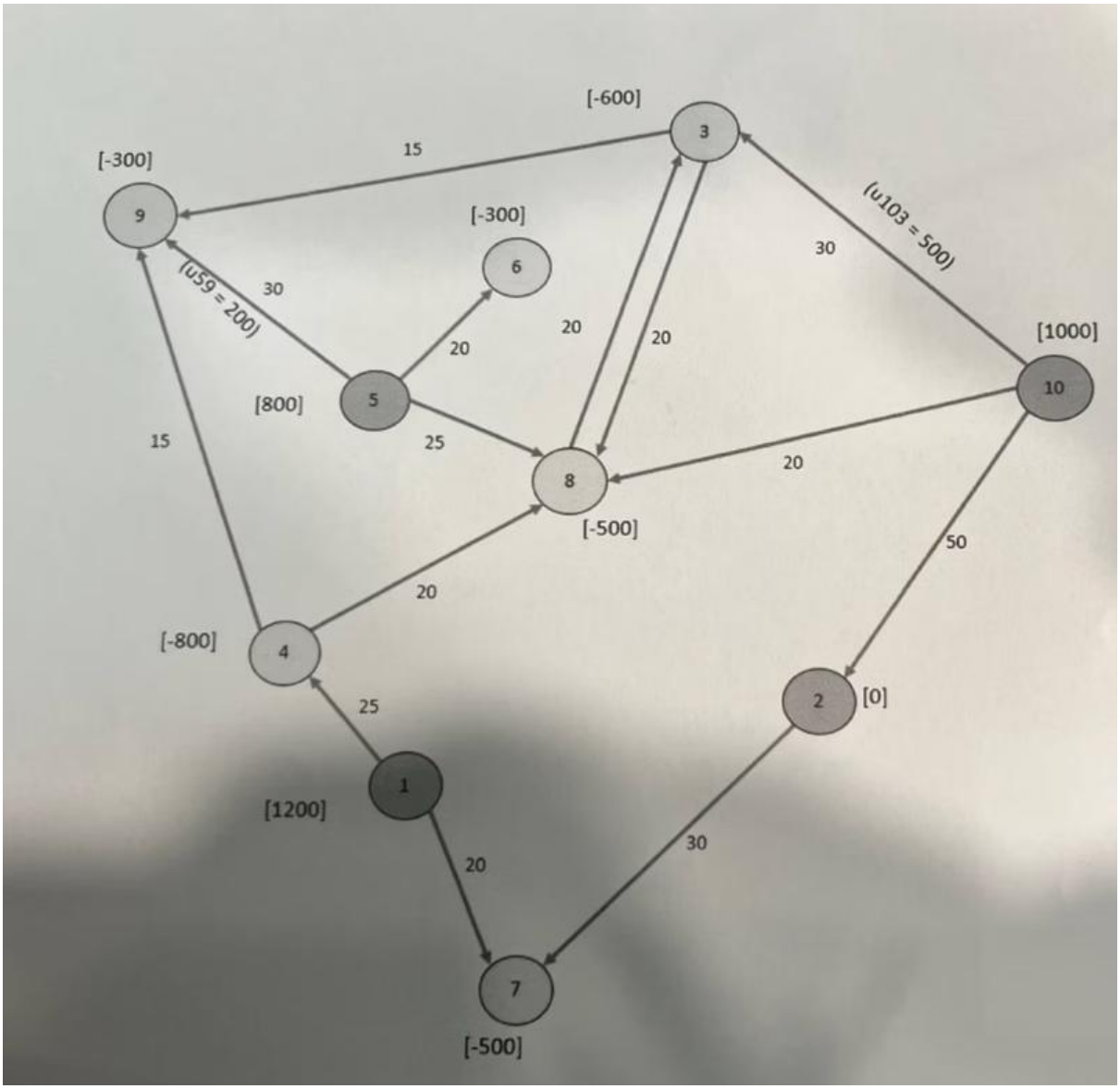
Work out a distribution plan that meets the demand for its educational toys and minimize the total cost of shipment.

1. Based on the information above, complete the table below.

Node	Location	Type of Node (Supply, Transhipment or Demand)	Net flow (number of units; Outflow is +ve, Inflow is -ve)
1	**Singapore**	**Supply**	**1200**
2	Bandar Seri Begawan	Transhipment	0
3	Hanoi	Demand	-600
4	Kuala Lumpur	Demand	-800
5	**Bangkok**	**Supply**	**800**
6	Vientiane	Demand	-300
7	Jakarta	Demand	-500
8	Phnom Penh	Demand	-500
9	Naypyidaw	Demand	-300
10	**Manila**	**Supply**	**1000**

2. What is the function of transhipment? How is Bandar Seri Begawan different from the other nodes?
   Goods are shipped from known supply points (manufacturing centers) to known demand points (customers), possibly through intermediate points (regional and or/field warehouses). It must satisfy the conservation of flow, so that flow in **equals** flow out.

3. Draw a network representation of the routes available to WantTheToys to distribute their toys.



4. The distribution flow problem faced by WantTheToys is known as a **minimum cost flow problem**. Similar to the transportation problem, it has the 'integer solution property' and can be formulated and solved using a linear programming (LP) model.

   a. Define the decision variables for this LP model. In total, how many decision variables are there?

   Decision variables: Xij represents the amount of flow through arc i to j where i, j represents nodes in the network:

   X14, X17, X27, X38, X39, X48, X49, X56, X58, X59, X83, X102, X103, X108

   There are 14 decision variables.

Model answer (MC HoweChiat) from SCORM: Let X_{ij} be the *number of units of toys (in '000s)* to be shipped from node i to node j. In total, there are 14 decision variables:
X_{14}, X_{17}, X_{27}, X_{38}, X_{39}, X_{48}, X_{49}, X_{56}, X_{58}, X_{59}, X_{83}, X_{102}, X_{103}, X_{108}

b. Write down the objective function of the minimum cost flow problem using the decision variables defined above.
Minimize total shipment cost, Z = 25X14 + 20X17 + 30X27 + 20X38 + 15X39 + 20X48 + 15X49 + 20X56 + 25X58 + 30X59 + 20X83 + 50X102 + 30X103 + 20X108

c. There are <u>three types of constraints</u> in the minimum cost flow LP model.

i. Formulate the **Net flow constraint** for each node in the network. This type of constraint has the following form:

Amount shipped out – Amount shipped in = Net flow generated at the node

Node 1: X14 + X17 = 1200
Node 2: X27 – X102 = 0
Node 3: X38 + X39 – X83 – X103 = -600
Node 4: X48 + X49 – X14 = -800
Node 5: X56 + X58 + X59 = 800
Node 6: -X56 = -300
Node 7: -X17 – X27 = -500
Node 8: X83 – X38 – X58 - X108 = -500
Node 9: -X39 – X49 – X59 = -300
Node 10: X102 + X103 + X108 = 1000

ii. Formulate the **Upper-bound constraints** for arcs having limited shipping capacities.
Furthermore, there is a shipping capacity limit of 500 from Manila to Hanoi and 200 from Bangkok to Naypyidaw.
Arc 10 to 3: X103 <= 500
Arc 5 to 9: X59 <= 200

iii. Formulate the **Non-negativity constraint** for each decision variable.
All Xij >= 0

5. Construct the minimum cost flow LP model in Excel Solver (template attached below). Build two models – one without the arc capacity constraints and one with arc capacity constraints. Compare and comment on their solutions.

AY 2025 Sem 2 E211
L20 Template.xlsx

Minimize Total Cost:
Without arc capacity constraints = 80000 (Amt flow = 3100)
With arc capacity constraints = 84000 (Amt flow = 3400)

6. Explain how the constraints in the minimum cost flow model should change for each of the following cases. Obtain a new optimal solution for each case.

   a. The plant in Singapore (Node 1) experienced a disruption and can only supply 1000 units of toys instead of 1200 **(Ans: 79000)**

When Total Supply (2800) < Total Demand (3000),

- This means that not all demand will be fulfilled.
- All demand node sign become “>=” (because RHS value is negative)

Decision Variables	Amount Flow	Unit Cost
X14	600	25
X17	400	20
X27	100	30
X38	0	20
X39	100	15
X48	0	20
X49	0	15
X56	300	20
X58	300	25
X59	200	30
X83	200	20
X102	100	50
X103	500	30
X108	400	20

Minimize Total Cost = 79000

Net Flow Constraints	constraint equation LHS	LHS	Sign	RHS
Supply Node 1:	X14+X17=1000	1000	=	1000
Supply Node 5:	X56+X58+X59=800	800	=	800
Supply Node 10:	X102+X103+X108=1000	1000	=	1000
Transhipment Node 2:	X27-X102=0	0	=	0
Demand Node 3	X38+X39-X83-X103=-600	-600	>=	-600
Demand Node 4	X48+X49-X14=-800	-600	>=	-800
Demand Node 6:	-X56=-300	-300	>=	-300
Demand Node 7:	-X17-X27=-500	-500	>=	-500
Demand Node 8	X83-X38-X58-X108=-500	-500	>=	-500
Demand Node 9:	-X39-X49-X59=-300	-300	>=	-300
Arc Capacity Constraints				
Arc 10-3	X103 <= 500	500	<=	500
Arc 5-9	X59 <= 200	200	<=	200

b. The demand at Phnom Penh is reduced to 400 **(Ans: 81500)**

When Total Supply (3000) > Total Demand (2900),

- This means that not all supply need to be sent out.
- All supply node sign becomes "<="

Decision Variables	Amount Flow	Unit Cost
X14	800	25
X17	400	20
X27	100	30
X38	0	20
X39	100	15
X48	0	20
X49	0	15
X56	300	20
X58	200	25
X59	200	30
X83	200	20
X102	100	50
X103	500	30
X108	400	20

Minimize Total Cost = 81500

Net Flow Constraints	constraint equation LHS	LHS	Sign	RHS
Supply Node 1:	X14+X17=1200	1200	<=	1200
Supply Node 5:	X56+X58+X59=800	700	<=	800
Supply Node 10:	X102+X103+X108=1000	1000	<=	1000
Transhipment Node 2:	X27-X102=0	0	=	0
Demand Node 3	X38+X39-X83-X103=-600	-600	=	-600
Demand Node 4	X48+X49-X14=-800	-800	=	-800
Demand Node 6:	-X56=-300	-300	=	-300
Demand Node 7:	-X17-X27=-500	-500	=	-500
Demand Node 8	X83-X38-X58-X108=-500	-400	=	-400
Demand Node 9:	-X39-X49-X59=-300	-300	=	-300
Arc Capacity Constraints				
Arc 10-3	X103 <= 500	500	<=	500
Arc 5-9	X59 <= 200	200	<=	200

School of Engineering

REPUBLIC POLYTECHNIC

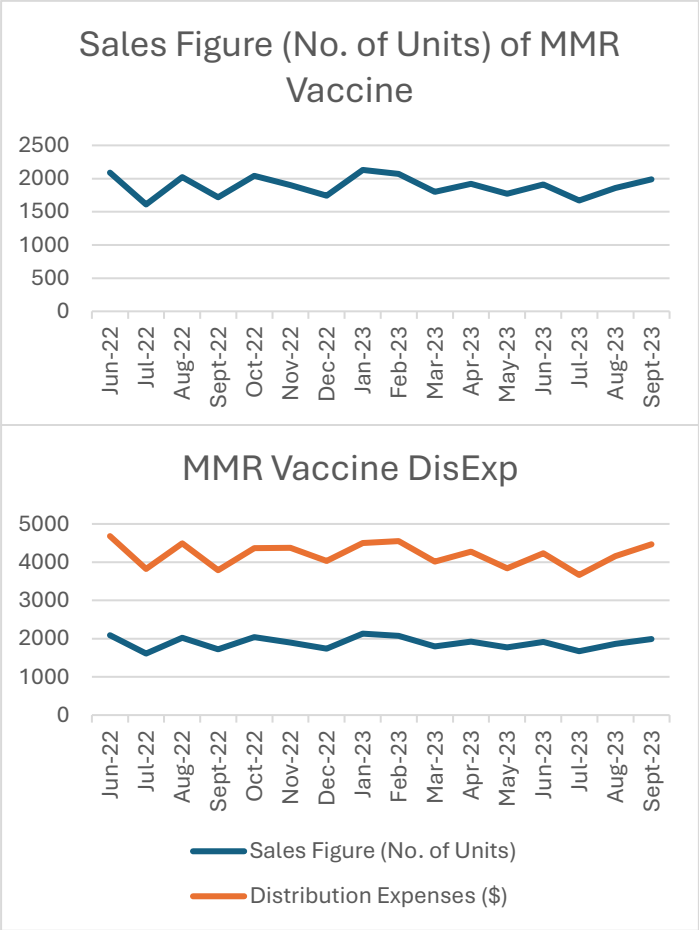
E211 – Operations Planning II

Worksheet for Lesson 26: Demand Forecasting

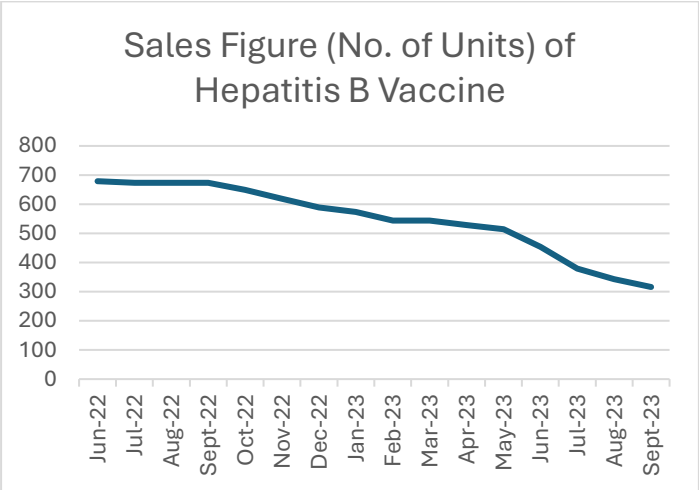
AY 2025 Sem 2 L26 Forecasting Data.xlsx

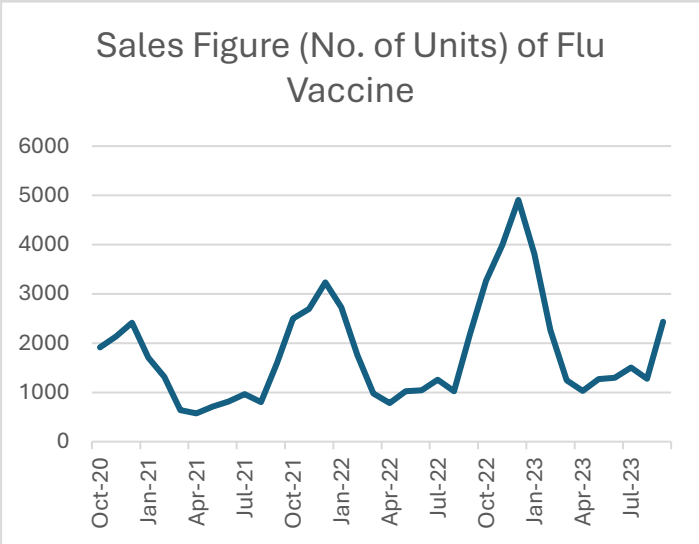
1. For each data set in the above Excel file, (Ashikeen)

   a) Plot a relevant graph



School of Engineering





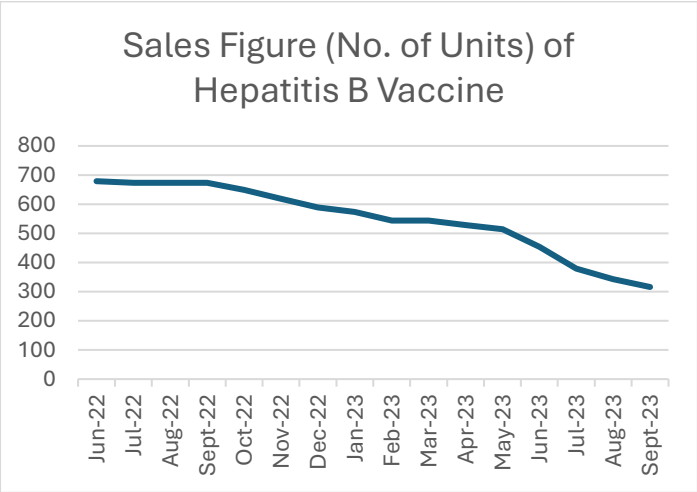
b) Identify the time series components present
   1. Random
   2. Random
   3. Trend
   4. Seasonality

c) Recommend a suitable forecasting method
   1. Moving Average
   2. Weighted Moving Average
   3. Holt's Method
   4. Multiplicative decomposition

Causal Forecasting – Simple Linear Regression

Which data set do you recommend using linear regression as the forecasting method?

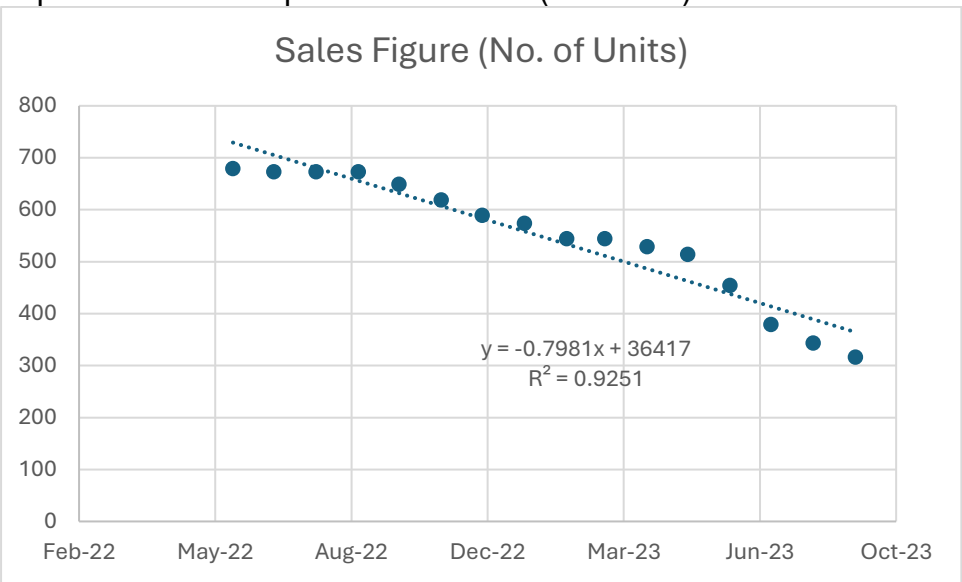
School of Engineering



2. For this data set, identify the dependent variable and the independent variable. (Ashikeen)

   Dependent Variable: Time
   Independent Variable: Sales figures

3. Use the scatter plot you have plotted to investigate the relationship between the dependent and independent variable. (Ashikeen)



   a) The relationship between the dependent variable and the independent variable? (Hint: linear or non-linear? Positive or negatively correlated?)
   There is a negative, linear relationship between time and the sales units of the Hepatitis B vaccine.

   b) Plot a linear regression line. The equation of the data's trend line is -0.9251.

   $\beta_0 = -0.7891$, $\beta_1 = 36417$

4. Obtain and interpret the following regression model statistics:
   a) Correlation coefficient, r -SQRT(0.9251) = -0.96182
   b) Coefficient of determination, R^2 ('R Square') 0.9251

5. Using the linear regression model, predict the dependent variable if the independent variable has a value of 2500.
   Dependent variable (y), Independent variable (x)
   Y = -0.7981x + 36417
   = -0.7981*(2500) + 36417
   = 34421.75

Time Series Forecasting – Moving Average and Exponential Smoothing [Celest]

Which data set do you recommend using moving average and exponential smoothing as the forecasting methods? MMR Vaccine

6. Moving Average is commonly used to derive forecast for time series with random/irregular variations. For the data set under study, develop a N-period moving average model that forecasts the monthly sales. Given N is the number of averaging periods,

   a) Use N = 3 (You can assume other values for N) to forecast the monthly sales.

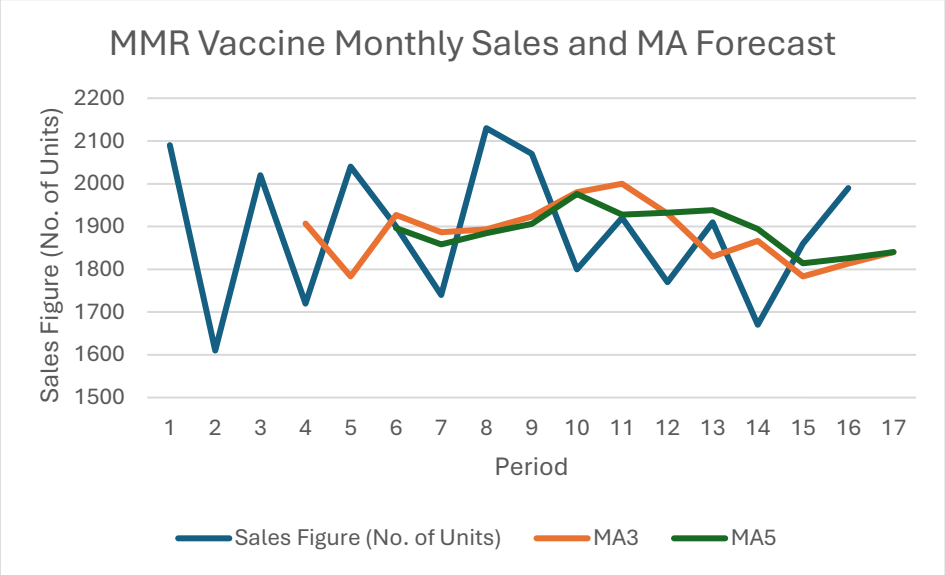
Month	Sales Figure (No. of Units)	MA3
Jun-22	2090	
Jul-22	1610	
Aug-22	2020	
Sep-22	1720	1906.67
Oct-22	2040	1783.33
Nov-22	1900	1926.67
Dec-22	1740	1886.67
Jan-23	2130	1893.33
Feb-23	2070	1923.33
Mar-23	1800	1980
Apr-23	1920	2000
May-23	1770	1930
Jun-23	1910	1830
Jul-23	1670	1866.67
Aug-23	1860	1783.33
Sep-23	1990	1813.33
		1840

   b) Use N = 5 (You can assume other values for N)) to forecast the monthly sales.

Month	Sales Figure (No. of Units)	MA3	MA5
Jun-22	2090		
Jul-22	1610		
Aug-22	2020		
Sep-22	1720	1906.67	
Oct-22	2040	1783.33	
Nov-22	1900	1926.67	1896
Dec-22	1740	1886.67	1858
Jan-23	2130	1893.33	1884
Feb-23	2070	1923.33	1906
Mar-23	1800	1980	1976
Apr-23	1920	2000	1928
May-23	1770	1930	1932
Jun-23	1910	1830	1938
Jul-23	1670	1866.67	1894
Aug-23	1860	1783.33	1814
Sep-23	1990	1813.33	1826
		1840	1840

c) Calculate Mean Absolute Deviation (MAD) and Mean Absolute Percent Error (MAPE) for a) and b).
MA3 → MAD = 150.00, MAPE = 7.97%
MA5 → MAD = 121.82, MAPE = 6.49%

d) Plot the forecasted values from a) and b) with the actual sales figures and comment on the graphs.



N=3 MA graph is more responsive to changes in sales figures compared with N=5 MA graph.

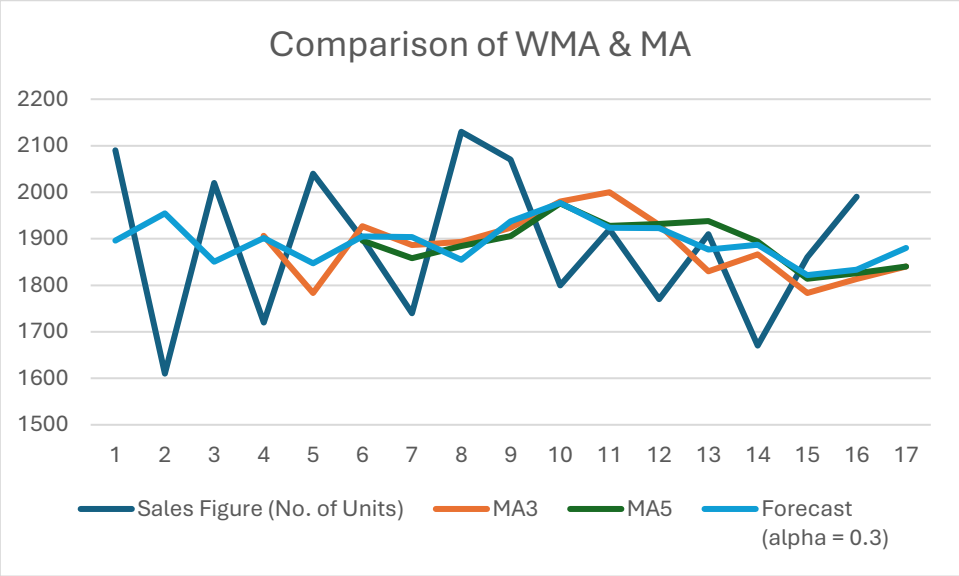
e) What happens to the forecast when you increase N?
As N increases, the forecast becomes less responsive to changes in actual data and the graph becomes smoother.

7. For the forecasts in Q6, the underlying assumption is that all data points are of equal importance (equal weights).

a) Assign weights (You can assume values for the weights) to the 3-period moving average to predict the monthly sales.
Weight 5,3,1

b) Calculate Mean Absolute Deviation (MAD) and Mean Absolute Percent Error (MAPE) for the weighted moving average forecasts.
MAD = 151.88, MAPE = 0.08

c) Plot the forecasted values to the moving average graphs and compare the use of **weighted moving average** and **moving average** in forecasting.

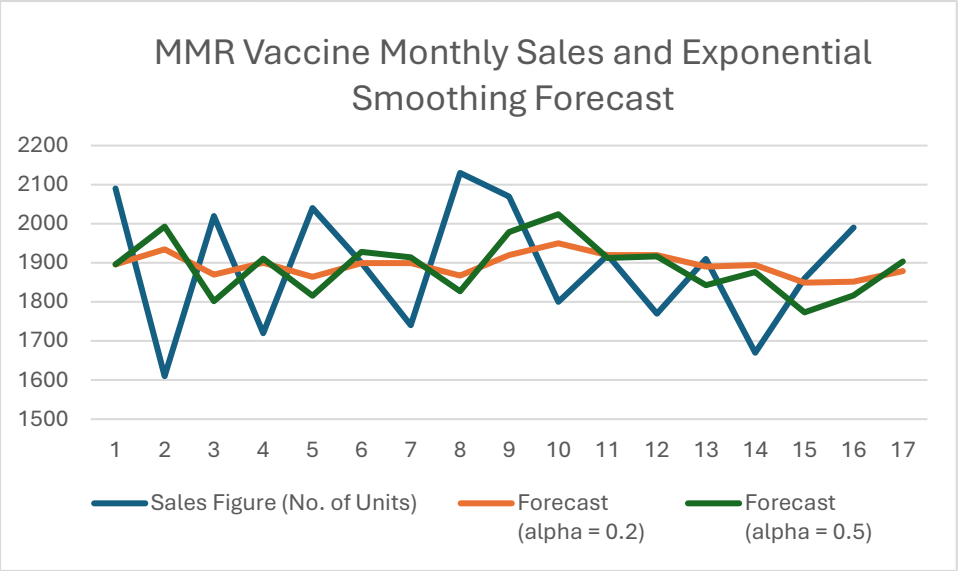


WMA is more responsive to current trends compared with moving average, which weighs all points equally.

8. Another commonly used forecasting method for time series with random/irregular variations is simple exponential smoothing. Develop a **simple exponential smoothing** model for the same data set to forecast the monthly sales. Given that α (Alpha) is the smoothing constant,

a) Use α = 0.2 (You can assume other values for α) to predict the monthly sales.
b) Use α = 0.5 (You can assume other values for α) to predict the monthly sales.
c) Calculate Mean Absolute Deviation (MAD) and Mean Absolute Percent Error (MAPE) for a) and b).
MAD (when α = 0.2) = 143.20, MAPE = 7.72%
MAD (when α = 0.5) = 169.92, MAPE = 9.15%

d) Plot the forecasted values from a) and b) with the actual sales figures and comment on the graphs. What happens to the forecast when you vary *α*?



α =0.5 SES is more responsive to changes in actual sales compared with α =0.2 SES. As α decreases, the forecast becomes less responsive to changes in actual data and the graph becomes smoother.

9. Tabulate and compare the forecasting errors for the methods applied in Q6 to Q8. Which forecasting method is most suitable for forecasting sales in Oct-23?

Forecasting methods	**Measures of forecasting error**	
	MAD	MAPE
MA (3-Period)	150.00	7.97%
MA (5-Period)	121.82	6.49%
WMA (3-Period)	151.88	8.08%
SES (0.2)	143.20	7.72%
SES (0.5)	169.92	9.15%

Time Series Forecasting – Trend Analysis using Linear Regression and Holt's Method

Which data set do you recommend using Holt's method as the forecasting method? MMR Vaccine-DisExp

10. Perform Linear Regression to forecast the monthly sales of this data set.

School of Engineering

x (independent variable)	y (dependent variable)		
Distribution Expenses ($)	Sales Figure (No. of Units)	Forecasted Sales Prices	AD
4682	2090	2121.107	31.107
3820	1610	1704.33	94.33
4496	2020	2031.176	11.176
3793	1720	1691.2755	28.7245
4367	2040	1968.8045	71.1955
4375	1900	1972.6725	72.6725
4028	1740	1804.898	64.898
4504	2130	2035.044	94.956
4548	2070	2056.318	13.682
4015	1800	1798.6125	1.3875
4276	1920	1924.806	4.806
3837	1770	1712.5495	57.4505
4237	1910	1905.9495	4.0505
3666	1670	1629.871	40.129
4156	1860	1866.786	6.786
4468	1990	2017.638	27.638
		MAD	39.0618125

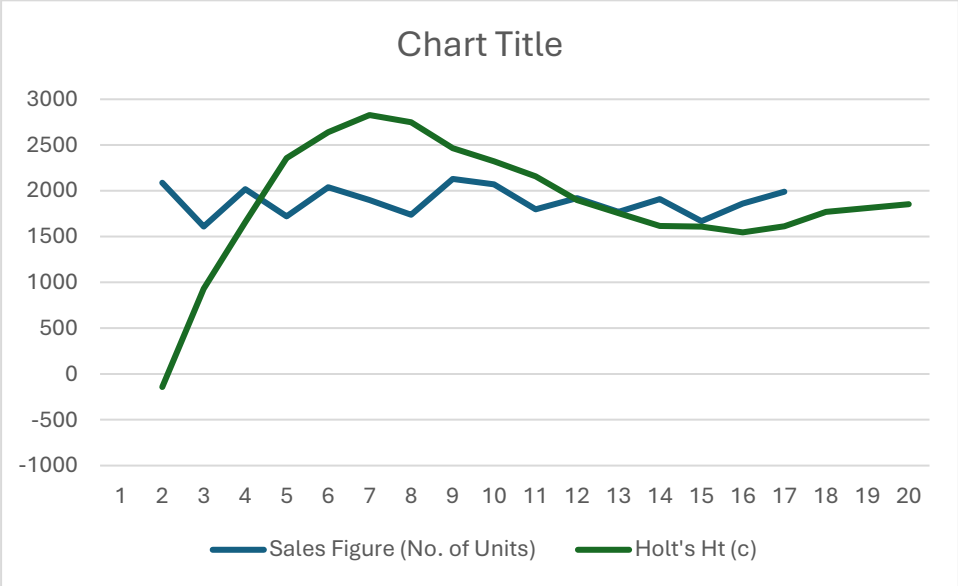
$y = 0.4835x – 142.64$

Sub x (distribution expenses) to get the forecasted sales prices respectively.

11. Based on Holt's method, determine the forecasts up to Dec-23. Assume α (for level estimation) = 0.3 and β (for trend estimation) = 0.6 (You can assume other values for the smoothing constants α and β).

    a) What values would you use from your regression analysis to represent L0 and T0?
    L0 = -142.6375, T0 = 0.4835

    b) Plot the forecasted values with the actual sales figures and comment on the graphs.



12. Compare the MAD and MAPE of Holt's method and Linear Regression forecasts. Which method would you recommend deriving forecasts for the monthly sales of this data set? Explain.
    I will recommend Holt's method as it gives a more accurate forecast due to having smaller forecasting errors.

    Holt's method:

MAD	529.5241	
MAPE		27.79%

    Linear Regression:

MAD	2028.53	
MAPE		107.38%

Time Series Forecasting – Handling Trend and Seasonality using Decomposition Method

Which data set do you recommend using Multiplicative Decomposition Method as the forecasting method? Flu Vaccine

15. There are two general forms of time-series decomposition models: multiplicative and additive. Which model is more suitable to be used to forecast the monthly sales of the data set? Justify your answer.
    Multiplicative model is more suitable to be used to forecast the monthly sales. As the seasonal trend increases every cycle rather than staying constant.

16. Write down the decomposition model identified in Q15 that Andy can use to forecast the monthly sales based on the four time series components.

    Let Y_t be the observed value of the time series in time t;
    $\mathbf{T}_t$ be the trend component in time t;
    $\mathbf{S}_t$ be the seasonal component in time t;
    $\mathbf{C}_t$ be the cyclic component in time t; and
    $\mathbf{I}_t$ be the irregular (or random) component in time t.

17. In the decomposition method, Moving Averages (MA) and/or Centred Moving Averages (CMA) are often used to eliminate seasonal variations and irregular fluctuations from the data. Answer the following questions:

    a) To remove the seasonal effect, what value should <u>the number of averaging periods</u> (N) be in MA calculation?
    N = 12

    b) Write down the formula to calculate the MA at Period 6 (Mar-21). Calculate the rest of the moving averages for this data set.

School of Engineering

REPUBLIC POLYTECHNIC

Mar-21 = AVERAGE(B3:B14)

Rest of MA for this data set:

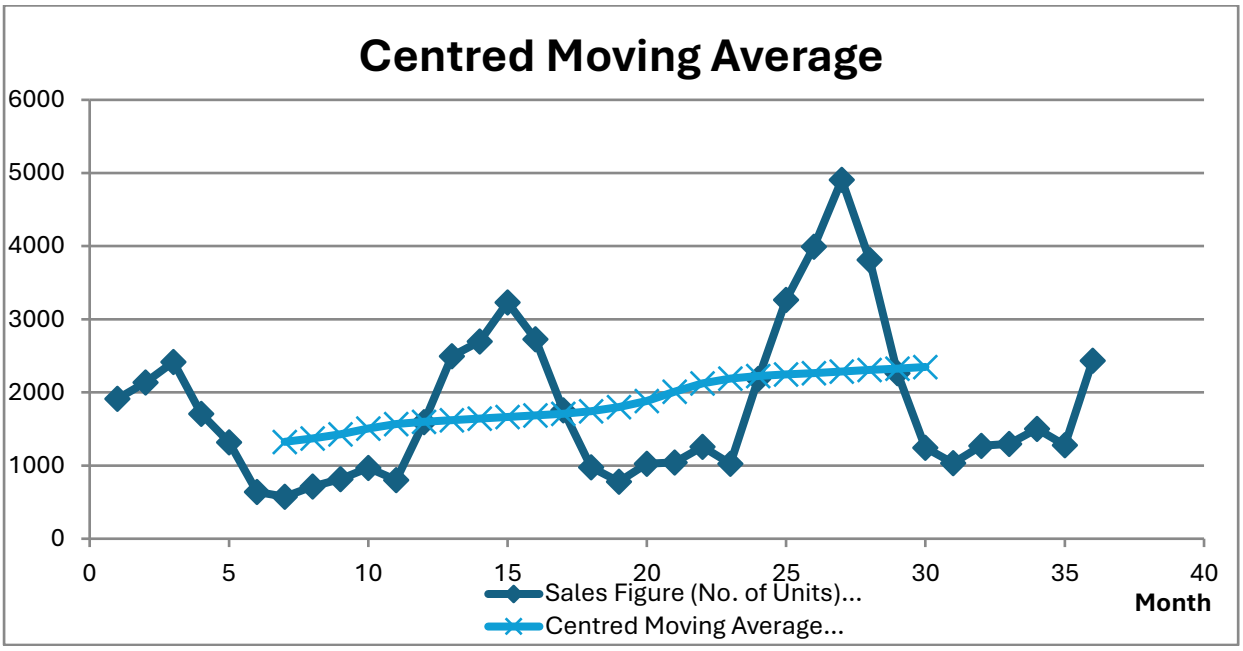
Month	Sales Figure (No. of Units) (Y)	Time Period (X)	Moving Average (e.g. 12 months)
Oct-20	1915	1	
Nov-20	2136	2	
Dec-20	2417	3	
Jan-21	1709	4	
Feb-21	1319	5	
Mar-21	639	6	1300.417
Apr-21	574	7	1348.833
May-21	713	8	1395.500
Jun-21	816	9	1463.333
Jul-21	967	10	1548.167
Aug-21	804	11	1584.500
Sept-21	1596	12	1612.667
Oct-21	2496	13	1629.917
Nov-21	2696	14	1656.000
Dec-21	3231	15	1675.083
Jan-22	2727	16	1699.250
Feb-22	1755	17	1717.667
Mar-22	977	18	1766.833
Apr-22	781	19	1831.000
May-22	1026	20	1939.167
Jun-22	1045	21	2078.833
Jul-22	1257	22	2169.250
Aug-22	1025	23	2211.833
Sept-22	2186	24	2234.000
Oct-22	3266	25	2255.000
Nov-22	3994	26	2275.417
Dec-22	4907	27	2296.250
Jan-23	3812	28	2316.667
Feb-23	2266	29	2337.833
Mar-23	1243	30	2358.750
Apr-23	1033	31	
May-23	1271	32	
Jun-23	1295	33	
Jul-23	1502	34	
Aug-23	1279	35	
Sept-23	2437	36	

c) When and why do you need to calculate the CMA? How is it different from MA?
CMA is calculated only when N is an even number to remove the seasonal variations from data.

d) Calculate the CMA for Period 7 (Apr-21) onwards.

Month	Sales Figure (No. of Units) (Y)	Time Period (X)	Moving Average (e.g. 12 months)	Centred Moving Average (=(MA_{i-1} + MA_i)/2)
Apr-21	574	7	1348.833	1324.625
May-21	713	8	1395.500	1372.167
Jun-21	816	9	1463.333	1429.417
Jul-21	967	10	1548.167	1505.750
Aug-21	804	11	1584.500	1566.333
Sept-21	1596	12	1612.667	1598.583
Oct-21	2496	13	1629.917	1621.292
Nov-21	2696	14	1656.000	1642.958
Dec-21	3231	15	1675.083	1665.542
Jan-22	2727	16	1699.250	1687.167
Feb-22	1755	17	1717.667	1708.458
Mar-22	977	18	1766.833	1742.250
Apr-22	781	19	1831.000	1798.917
May-22	1026	20	1939.167	1885.083
Jun-22	1045	21	2078.833	2009.000
Jul-22	1257	22	2169.250	2124.042
Aug-22	1025	23	2211.833	2190.542
Sept-22	2186	24	2234.000	2222.917
Oct-22	3266	25	2255.000	2244.500
Nov-22	3994	26	2275.417	2265.208
Dec-22	4907	27	2296.250	2285.833
Jan-23	3812	28	2316.667	2306.458
Feb-23	2266	29	2337.833	2327.250
Mar-23	1243	30	2358.750	2348.292

18. Plot a graph for the CMA and the actual values, what do you observe?



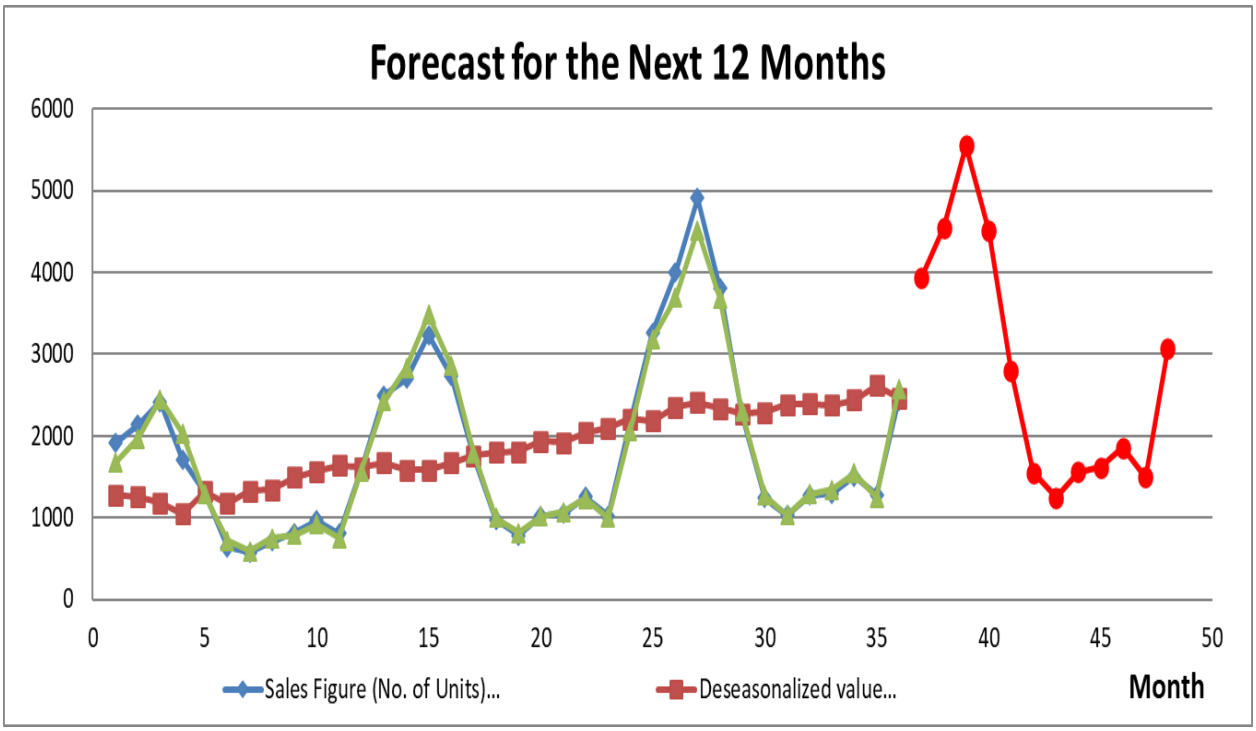
19. Produce the forecasts for the next 12 months from Oct 2023 by working out the following in order:

    a) Seasonal ratio
    b) Seasonal index
    c) Deseasonalized value
    d) Deseasonalized forecast
    e) Seasonalized forecast

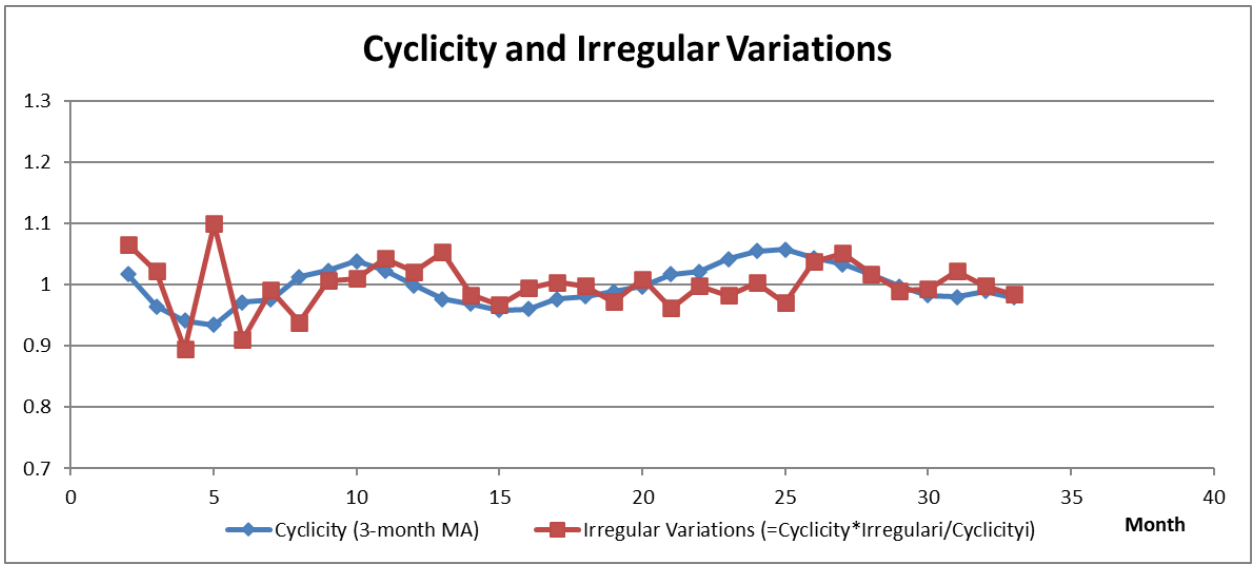
School of Engineering

Seasonal Ratio (=Y_i/CMA_i)	Seasonal Index (= normalized average of seasonal ratio)	Deseasonalized value (= Y_i / Seasonal $Index_i$)	Deseasonalized forecast (=slope*X_i + intercept)	Seasonalized forecast (= Seasonal $Index_i$ *Deseasonalized $forecast_i$)
	1.493	1282.413	1118.658	1670.467
	1.697	1258.335	1160.905	1970.614
	2.038	1186.084	1203.151	2451.781
	1.630	1048.382	1245.398	2030.161
	0.998	1321.954	1287.645	1284.768
	0.544	1175.548	1329.891	722.898
0.433	0.433	1326.948	1372.138	593.548
0.520	0.531	1343.987	1414.385	750.347
0.571	0.544	1499.887	1456.632	792.467
0.642	0.615	1571.494	1498.878	922.317
0.513	0.489	1643.201	1541.125	754.055
0.998	0.988	1615.028	1583.372	1564.717
1.540	1.493	1671.490	1625.618	2427.501
1.641	1.697	1588.235	1667.865	2831.170
1.940	2.038	1585.534	1710.112	3484.864
1.616	1.630	1672.872	1752.359	2856.573
1.027	0.998	1758.930	1794.605	1790.596
0.561	0.544	1797.356	1836.852	998.469
0.434	0.433	1805.482	1879.099	812.844
0.544	0.531	1933.984	1921.345	1019.295
0.520	0.544	1920.812	1963.592	1068.274
0.592	0.615	2042.780	2005.839	1234.269
0.468	0.489	2094.878	2048.086	1002.105
0.983	0.988	2212.062	2090.332	2065.705
1.455	1.493	2187.134	2132.579	3184.534
1.763	1.697	2352.898	2174.826	3691.726
2.147	2.038	2407.990	2217.072	4517.948
1.653	1.630	2338.463	2259.319	3682.985
0.974	0.998	2271.074	2301.566	2296.424
0.529	0.544	2286.707	2343.813	1274.041
	0.433	2388.045	2386.059	1032.141
	0.531	2395.802	2428.306	1288.244
	0.544	2380.336	2470.553	1344.082
	0.615	2440.935	2512.799	1546.221
	0.489	2613.998	2555.046	1250.155
	0.988	2466.054	2597.293	2566.692

20. Plot the monthly sales forecasts for the next 12 months. Comment on the validity of the Multiplicative Decomposition Model built.



21. Investigate if there are significant cyclical components in the sales data by further decomposing the seasonal forecasts into cyclical and irregular components.
[Copilot]
There are no significant cyclical components in the sales data.
After removing trend and seasonal effects, the remaining cyclical–irregular series fluctuates randomly around 1.0 without any stable or repeating multi-year pattern. This indicates that the deviations are primarily due to irregular (random) variation rather than true cyclical behaviour.

22. Plot the cyclical and irregular components of the time series. Do they have significant effect on the forecasts? If under certain conditions, the cyclical component plays a significant role in the forecast, how would you incorporate it into the forecasting?

School of Engineering